

MTH166

Lecture-19

Euler-Cauchy Equation-II

Topic:

Solution of Homogeneous and Non-Homogeneous LDE with Variable coefficients.

Learning Outcomes:

To convert equation with variable coefficients (Euler-Cauchy Equation) to an equation with constant coefficients and then solve it with standard known methods.

Problem 1. Find the general solution of: $x^2 y'' - 2y = 2x^2$

Solution: The given equation is:

$$x^2 y'' - 2y = 2x^2 \quad (1)$$

$$\text{S.F.: } (x^2 D^2 - 2)y = 2x^2 \quad (2) \quad \text{where } D \equiv \frac{d}{dx}$$

Let transformation be: $x = e^t \Rightarrow t = \log x$

then $x D = \theta$, $x^2 D^2 = \theta(\theta - 1)$ where $\theta \equiv \frac{d}{dt}$

Equation (2) becomes: $[\theta(\theta - 1) - 2]y = 2e^{2t}$

$\Rightarrow f(\theta)y = r(t)$ where $f(\theta) = (\theta^2 - \theta - 2)$ and $r(t) = 2e^{2t}$

To find Complimentary Function (C.F.):

$$\text{A.E.: } f(\theta) = 0 \Rightarrow (\theta^2 - \theta - 2) = 0 \Rightarrow (\theta - 2)(\theta + 1) = 0$$

$\Rightarrow \theta = 2, -1$ (real and unequal roots)

$\therefore$ Complimentary function is given by:

$$\Rightarrow y_c = c_1 e^{2t} + c_2 e^{-t} \Rightarrow y_c = c_1 x^2 + c_2 x^{-1}$$

To find Particular Integral (P.I.):

$$y_p = \frac{1}{f(\theta)} r(t) = \frac{1}{(\theta^2 - \theta - 2)} (2e^{2t})$$

$$\Rightarrow y_p = 2 \left[\frac{1}{(\theta^2 - \theta - 2)} e^t \right] \Rightarrow y_p = 2 \left[\frac{1}{((2)^2 - (2) - 2)} e^t \right] \quad (\text{Put } \theta = 2)$$

$$\Rightarrow y_p = 2 \left[\frac{1}{4-4} e^t \right] \quad \text{Case of failure}$$

$$y_p = 2t \left[\frac{1}{2\theta - 1} e^{2t} \right] = 2t \left(\frac{1}{3} e^{2t} \right) = 2 \log x \left(\frac{1}{3} x^2 \right)$$

$\therefore$ General solution is given by: $y = \text{C.F.} + \text{P.I.}$

$$\text{i.e. } y = y_c + y_p$$

$$\Rightarrow y = (c_1 x^2 + c_2 x^{-1}) + 2 \log x \left(\frac{1}{3} x^2 \right) \textbf{Answer.}$$

Problem 2. Find the general solution of: $x^2 y'' + 2xy' = \cos(\log x)$

Solution: The given equation is:

$$x^2 y'' + 2xy' = \cos(\log x) \quad (1)$$

$$\text{S.F.: } (x^2 D^2 + 2xD)y = \cos(\log x) \quad (2) \quad \text{where } D \equiv \frac{d}{dx}$$

Let transformation be: $x = e^t \Rightarrow t = \log x$

$$\text{then } xD = \theta, \quad x^2 D^2 = \theta(\theta - 1) \quad \text{where } \theta \equiv \frac{d}{dt}$$

Equation (2) becomes: $[\theta(\theta - 1) + 2\theta]y = \cos t$

$$\Rightarrow f(\theta)y = r(t) \quad \text{where } f(\theta) = (\theta^2 + \theta) \quad \text{and } r(t) = \cos t$$

To find Complimentary Function (C.F.):

$$\text{A.E.: } f(\theta) = 0 \Rightarrow (\theta^2 + \theta) = 0 \Rightarrow \theta(\theta + 1) = 0$$

$$\Rightarrow \theta = 0, -1 \quad (\text{real and unequal roots})$$

$\therefore$ Complimentary function is given by:

$$\Rightarrow y_c = c_1 e^{0t} + c_2 e^{-t} \Rightarrow y_c = c_1 + c_2 x^{-1}$$

To find Particular Integral (P.I.):

$$y_p = \frac{1}{f(\theta)} r(t) = \frac{1}{(\theta^2 + \theta)} (\cos t)$$

$$\Rightarrow y_p = \left[\frac{1}{(-1)^2 + \theta} (\cos t) \right] \quad (\text{Put } \theta^2 = -(1)^2)$$

$$\Rightarrow y_p = \left[\frac{1}{(\theta - 1)} (\cos t) \right] = \left[\frac{1}{(\theta - 1)} \times \frac{(\theta + 1)}{(\theta + 1)} (\cos t) \right] = \left[\frac{(\theta + 1)}{(\theta^2 - 1)} (\cos t) \right]$$

$$\Rightarrow y_p = \left[\frac{(\theta + 1)}{(-1 - 1)} (\cos t) \right] = -\frac{1}{2} \left[\frac{d}{dt} (\cos t) + (\cos t) \right] = -\frac{1}{2} (-\sin t + \cos t)$$

$$\Rightarrow y_p = -\frac{1}{2} [-\sin(\log x) + \cos(\log x)]$$

$\therefore$ General solution is given by: $y = \text{C.F.} + \text{P.I.}$

$$\text{i.e. } y = y_c + y_p$$

$$\Rightarrow y = c_1 + c_2 x^{-1} - \frac{1}{2} [-\sin(\log x) + \cos(\log x)] \quad \textbf{Answer.}$$

Polling Quiz

Which of the following is not an Euler-Cauchy differential equation:

(A) $x^2 y'' + xy' - 4y = 0$

(B) $x^2 y'' - 2xy = 2x$

(C) $x^2 y'' - 3xy' + 3y = 2 + 3 \log x$

Problem 3. Find the general solution of: $x^2 y'' - 3xy' + 3y = 2 + 3 \log x$

Solution: The given equation is:

$$x^2 y'' - 3xy' + 3y = 2 + 3 \log x \quad (1)$$

$$\text{S.F.: } (x^2 D^2 - 3xD + 3)y = 2 + 3 \log x \quad (2) \quad \text{where } D \equiv \frac{d}{dx}$$

Let transformation be: $x = e^t \Rightarrow t = \log x$

then $xD = \theta$, $x^2 D^2 = \theta(\theta - 1)$ where $\theta \equiv \frac{d}{dt}$

Equation (2) becomes: $[\theta(\theta - 1) - 3\theta + 3]y = 2 + 3t$

$\Rightarrow f(\theta)y = r(t)$ where $f(\theta) = (\theta^2 - 4\theta + 3)$ and $r(t) = 2 + 3t$

To find Complimentary Function (C.F.):

$$\text{A.E.: } f(\theta) = 0 \Rightarrow (\theta^2 - 4\theta + 3) = 0 \Rightarrow (\theta - 1)(\theta - 3) = 0$$

$\Rightarrow \theta = 1, 3$ (real and unequal roots)

$\therefore$ Complimentary function is given by:

$$\Rightarrow y_c = c_1 e^t + c_2 e^{3t} \Rightarrow y_c = c_1 x + c_2 x^3$$

To find Particular Integral (P.I.):

$$y_p = \frac{1}{f(\theta)} r(t) = \frac{1}{(\theta^2 - 4\theta + 3)} (2 + 3t)$$

$$\Rightarrow y_p = \left[\frac{1}{3 \left[1 + \left(\frac{\theta^2 - 4\theta}{3} \right) \right]} (2 + 3t) \right] = \frac{1}{3} \left[\left(1 + \left(\frac{\theta^2 - 4\theta}{3} \right) \right)^{-1} (2 + 3t) \right]$$

$$\Rightarrow y_p = \frac{1}{3} \left[\left(1 - \left(\frac{\theta^2 - 4\theta}{3} \right) + \left(\frac{\theta^2 - 4\theta}{3} \right)^2 - \dots \right) (2 + 3t) \right]$$

$$\Rightarrow y_p = \frac{1}{3} \left[(2 + 3t) + \frac{4}{3} \frac{d}{dt} (2 + 3t) + 0 \right] = \frac{1}{3} [(2 + 3t) + 4] = 2 + t$$

$$\Rightarrow y_p = 2 + \log x$$

$\therefore$ General solution is given by: $y = \text{C.F.} + \text{P.I.}$

$$\text{i.e. } y = y_c + y_p$$

$$\Rightarrow y = c_1 x + c_2 x^3 + 2 + \log x \quad \textbf{Answer.}$$

Q27. The Particular Integral (P.I.) for the differential equation $y'' + y' - 6y = 5e^{-3x}$ is:

(a) $-e^{-3x}$

(b) $-xe^{-3x}$

(c) xe^{-3x}

(d) e^{-3x}

Q28. By method of undetermined coefficient, the choice of particular integral for $y'' - 4y = 5e^{-2x}$ is

(a) $C e^{-2x}$

(b) $C x e^{-2x}$

(c) $C x^2 e^{-2x}$

(d) $C x^3 e^{-2x}$

Q29. To transform $xy'' + y' = \frac{1}{x}$ into a linear differential equations with constant coefficient, suitable transformation of x is

- (a) $x = \sin t$ (b) $x = \log t$ (c) $x = e^t$ (d) none of these

Q30. Which of the following is Euler Cauchy equation?

(a) $x^2y''-5xy'+13y=30xy^2$

(b) $x^2y''-5xy'+13xy=30x^2$

(c) $x^2y''-5xy'+13y=30xy^2$

(d) None of these

Q32. The P.I. of the differential equation $(D^3 + 6D + 9)y = 5e^{-x}$ is

(a) $\frac{e^{-x}}{2}$

(b) $\frac{5e^{-x}}{2}$

(c) $-\frac{e^{-x}}{2}$

(d) $-\frac{5e^{-x}}{2}$

